

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of Bachelor in Medical Imaging Technology Examination May 2018

BMRIT 203 Conventional Radiography and Equipment

Date: 26/05/2018

Day: Saturday

Time: 10:00am To 10:15pm

Marks: 10

Student's ID No.									
Student's Name: _____									
Student's Sign : _____									
Supervisor's Name : _____									
Supervisor's Sign: _____									

Answer Sheet for Multiple Choice Questions (MCQs), True or false and Match the following
Examination : BMIT/BOTAT/BOP
Semester: II
Section :
Course Code : _____
Course Name: _____

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		06
1.	The purpose of placing mA meter in x ray circuit is to know the		
	A.Current passing through the Filament circuit.	B. Current passing through the x ray tube.	
	C. Current passing through the Cathode of x ray tube.	D. Current passing through the Auto transformer.	
2.	The wavelength of X ray radiation generally used in x ray is		
	A. Longer than the visible light.	B. Longer than the UV rays.	
	C.longer than the gamma Radiation.	D.Shorter than the UV rays.	
3.	The process by which AC is converted into DC is known as		
	A.Transformer.	B. Rectification.	
	C.Thermoionic emission.	D. Generation.	

4.	The tendency to limit the emission of more electron from the filament of x ray tube during heating is known as		
	A.Thermoionic emission.	B. Space charge effect.	
	C.Voltage fluctuation	D. All of above	
5.	The phosphor employed in employed in conventional fluoroscopic screen is		
	A.Calcium tungstate	B. Zinc cadmium sulphide	
	C.Csium iodide	D .Barium sulphate	
6.	Full rectification system produces		
	A .Low tube rating	B. X rays on both cycle	
	C. Increase in exposure time	D. All of above	

Q – II	Select whether following statement is true or false.	4
		True/False
07.	The disadvantage of moving grid is longer exposure time and films get blurred.	
08.	First used x ray tube in conventional x ray tube is Coolidge tube	
09.	Small focal spot results better spatial Resolution.	
10.	Increase in kvp decrease photoelectric absorption and increase Compton scattering.	

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Date: 26/05/2018

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Time: 10:15am To 12:00pm

Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I		
Q - III Answer the following questions as directed		10
1	Explain working photomultiplier tube.	02
2	Explain line focus principle and heel effect compensation.	02
3	Explain construction of Collimator along with its advantages.	02
4	Explain working principle of fluoroscopy and list out its components.	02
5	Diagram of three phase 12 pulse generator.	02
SECTION - II		
Q-IV Answer the following questions as directed		15
1	List out types of x ray tube and describe modern x ray tube with diagram.	3
2	Explain image intensifier with a well labelled diagram.	3
3	Enlist the types of Exposure Timer and explain working principle of Cutie Pie.	3
4	Explain interaction of electron with matter	3
5	Draw a well labelled diagram of x ray circuit and add note on output from different types of Generator.	3